

2NF: For a table to be in the 2nd Normal Form... it must follow 2 conditions.

- it should be in 1st Normal form.

- And, it should not have any Partial Dependencies in the table.

→ What is Partial Dependency?
To understand partial Dependency, we first need to understand what Dependency is?

ex.

Student Table

Student id	name	reg-no	branch	address
1	Akon	CSE-18	CSE	TN
2	Akon	IT-18	IT	AP
3	Bkon	CSE-18	CSE	HR
4	Ckon	CSE-18	CSE	MH

→ A primary key: it is nothing but a column or a type or group of columns, which can uniquely identify each row in the table.

Note in above table Student id is primary key.

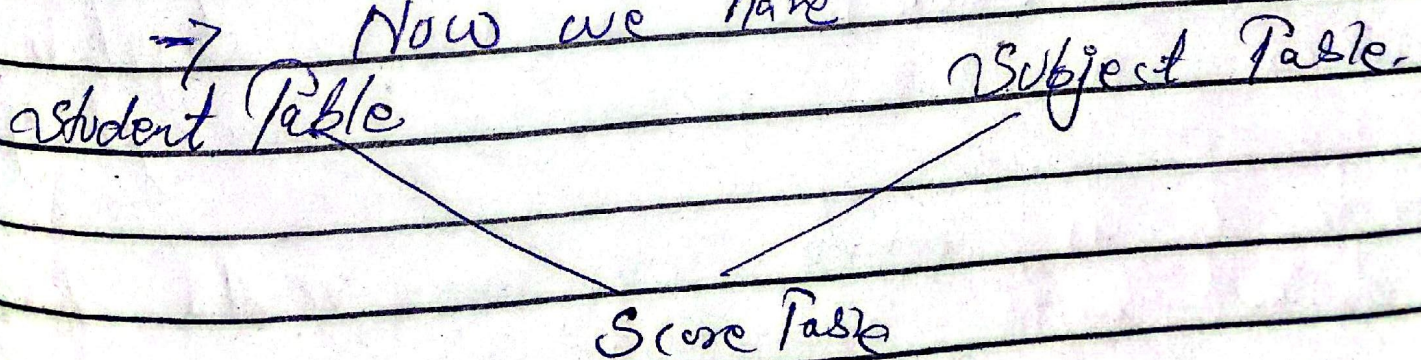
also for a simple table like a student
a single column which is studentid,
can uniquely identify the complete row
or all the other columns in a row

But this is not true all the time.
Now let's extend our ex to see
if two or more columns together can
act as a primary key or not.
let's create another table for subject.

in this table, we will have

- subject-id
- subject-name

→ Now we have



Score Table

scoreid	studentid	subjid	marks	teacher
1	10	1	82	Mr. J
2	10	2	77	Mr. G
3	11	1	85	Mr. J
4	11	2	82	Mr. C
5	11	4	95	Mr. P

in the score Table we are using Student-id to know which students marks.
Subject-id to know which subject we marks are.

Primary key should be.
score-id

But student-id + Subject-id together makes a more meaningful primary key.

→ if I ask ~~we~~ to get me marks of student, with student id 10.

can we get it? no
because we don't know from which subject.

if I give subject-id we don't know for which subject.

Hence we need student-id + subject-id to uniquely identify any row/entry in score table.

because we have
student

subject

Many to Many relationship

Many to Many relationship means.
one student can opt for more than
one subject

and

one subject can be opted by more
than one student.

so teacher column is depend on
subject-id.

ex.

subject-id = 1 is for java.
subject-id = 2 is for C++

Primary Key is a composition of
two columns in score table.
which is student-id + subject-id.

so the teacher name is only depend on
subject. it has nothing to do with student-id
(this is partial dependency)

How to remove partial dependency?

Our objective is to remove teacher column from sec table.

To move teacher column to subject Table.

Subject Table

Subject-id	Subject-name	teacher
1	Jing	Mr. j
2	c++	Mr. c++
3	C++	Mr. C++
4	php	Mr. p.

Makes more here sense

Teacher Table

teacherid	teacher-name
1	Mr. j
2	Mr. c++
3	Mr. C++
4	Mr. p.

so this is how 2nd Normal form works